# Repository Structure Guide

Facial Recognition with Emotion and Liveliness | COS30082 Applied Machine Learning, Swinburne University of Technology

**Rule of thumb:** Frontend-GUI only shows things on screen. Backend-Core does all the thinking (models and logic).

```
Facial_Recognition_with_Emotion_and_Liveliness/
|-- main                  (final submission only)
|-- develop               (team works here)
|-- README.md
|-- Documents/            (brief, reports)
|-- Frontend-GUI/         SCREEN + CAMERA ONLY
|   |-- ui/               windows, dialogs, overlay
|   |-- camera/           webcam capture
|   `-- assets/           icons, fonts, styles
`-- Backend-Core/         MODELS + LOGIC ONLY
    |-- configs/          config.yaml (all settings)
    |-- core/             inference pipeline
    |-- training/         train + evaluate
    |-- weights/          trained models (<100 MB)
    |-- registry_data/    enrolled users (embeddings)
    |-- scripts/          CLI helper tools
    `-- tests/            automatic checks
```

**How one frame flows**

```
camera/ -> pipeline.py -> detector
  -> liveness -> recognizer + emotion
  -> result -> ui/ overlay
```

**Team rules**

- Work on **develop**; **main** is for submission only.
- Never commit the dataset (**data/**, ***.zip**); it lives on Google Drive.
- No GUI code in Backend-Core; no model code in Frontend-GUI.
- GUI talks to backend only via **core/pipeline.py**.
- Folders currently hold a README; the .py files shown in tables are planned.

## Frontend-GUI (screen and camera only)

| | | |
|---|---|---|
| **ui/** | Windows and widgets | register_dialog.py shows "USER ALREADY EXISTS" if the face is already enrolled |
| **camera/** | Reads the laptop webcam | Grabs a frame and passes it to the backend |
| **assets/** | Icons, fonts, stylesheets | logo.png |

## Backend-Core (models and logic only)

| | | |
|---|---|---|
| **configs/** | All tunable settings in one place | Change threshold: 0.5 in config.yaml; no code edits needed |
| **core/** | The pipeline, one file per job | liveness.py: real face or photo? recognizer.py: who is this? emotion.py: which emotion? registry.py: enrolled users |
| **training/** | Train and evaluate models | train_emotion.py trains the emotion model; evaluate.py plots ROC and EER |
| **weights/** | Trained model files, each under 100 MB | emotion_model.pt |
| **registry_data/** | Enrolled users (embeddings only, no photos) | users.json |
| **scripts/** | Command-line helper tools | extract_dataset.py unzips the Drive dataset into data/ |
| **tests/** | Automatic checks for core/ | Enrolling the same face twice must be rejected |